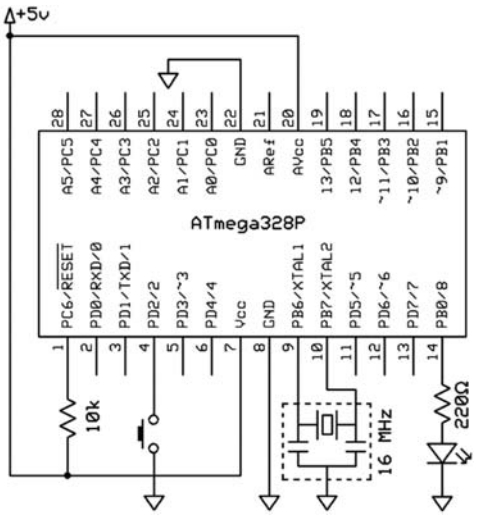
**Embedded Systems Development**

**Exercise 2**

1. Consider the following schematic diagram.




Write a firmware in assembly to blink the LED in every 200ms if the push button switch is pressed and otherwise in every 400ms.

2. Consider the following two programs written using Arduino IDE.

```
// Program 1
#include <avr/io.h>
int main(void)
{
  DDRB = 0x01;
  while (1)
  {
    PORTB = 0x01;
    PORTB = 0x00;
  }
  return 0;
}
```

```
// Program 2
void setup()
{
  pinMode(8, OUTPUT);
}

void loop()
{
  digitalWrite(8, HIGH);
  digitalWrite(8, LOW);
}
```

a. If the above two programs are compiled and uploaded to an ATmega328P microcontroller using Arduino IDE, what would be their intended functionality?

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b. It was observed that once compiled and uploaded to an ATmega328P microcontroller using Arduino IDE, the hex files of program 1 and program 2 consume 142 bytes and 743 bytes of microcontroller's code ROM, respectively. Explain this observation.

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3. If the following subroutine is assembled and uploaded into an Arduino Uno board, what is the time delay that it generates in milliseconds? Show all your workings clearly.

```
DELAY: LDI R16, 0x69
LOOP1: LDI R17, 0xFD
LOOP2: DEC R17
       BRNE LOOP2
       DEC R16
       NOP
       BRNE LOOP1
       RET
```

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